

lab 07

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Exercício 1

```
library(DBI)
library(RSQLite)

conn <- dbConnect(RSQLite::SQLite(), "database.sqlite3")
```

Exercício 2

```
query <- "
  SELECT DISTINCT i.id, i.name
  FROM subjects sub
  JOIN subject_memberships sm ON sub.code = sm.subject_code
  JOIN sections sec ON sm.course_offering_uuid = sec.course_offering_uuid
  JOIN teachings t ON sec.uuid = t.section_uuid
  JOIN instructors i ON t.instructor_id = i.id
  WHERE sub.abbreviation = 'STAT';
"

professores_stat <- dbGetQuery(conn, query)

# Quem são os professores
professores_stat$name
```

[1] "MINJING TAO"	"DONALD PORTER"
[3] "SHENG WANG"	"KUNLING HUANG"
[5] "DONG XIA"	"JENNIFER NGUYEN"
[7] "SEULKEE YUN"	"DUZHE WANG"

[9]	"YONGFENG WU"	"JARED BROWN"
[11]	"YUCHEN ZHOU"	"VICTOR LUO"
[13]	"GUN WOONG PARK"	"BROOK LUERS"
[15]	"DUY NGUYEN"	"JOHN GILLET"
[17]	"QUOC TRAN"	"YAOYAO XU"
[19]	"JIE ZHANG"	"LONG PHAN"
[21]	"WESLEY CHANG"	"TONG LI"
[23]	"SHANE HUBLER"	"FAN YANG"
[25]	"JU HEE CHO"	"JEEA CHOI"
[27]	"SHIXUE LIU"	"WEI-YIN LOH"
[29]	"BIN DAI"	"XIPEI YANG"
[31]	"DOUGLAS M. BATES"	"MICHAEL GEORGE ILTIS"
[33]	"LIAM JOHNSTON"	"CRYSTAL CHEN"
[35]	"YIFAN MEI"	"RUI CHEN"
[37]	"JILI WANG"	"LUWAN ZHANG"
[39]	"GINA BENNINGER"	"JIE SONG"
[41]	"XIWEN MA"	"HAODA FU"
[43]	"DEYUAN JIANG"	"JINGJIANG PENG"
[45]	"ERICA LEE DEADMAN"	"AKICHIKA OZEKI"
[47]	"MENG SONG"	"HAO TENG"
[49]	"DONGGYU KIM"	"COLIN LONGHURST"
[51]	"FREDERICK BOEHM"	"YONGSU LEE"
[53]	"GONZALO CONTADOR"	"HAO CHEN"
[55]	"XIN ZHANG"	"BO YANG"
[57]	"RUNGANG HAN"	"SEAN KENT"
[59]	"DEBRAJ DAS"	"YI LI"
[61]	"CLAIRE BOBST"	"CHAN PARK"
[63]	"SIJING LI"	"SHENGJI JIA"
[65]	"YUCHANG WU"	"ZHUANG WU"
[67]	"XIAOMAO LI"	"HYEBIN SONG"
[69]	"YOURAN QI"	"YOUNG MIN PARK"
[71]	"BIN ZHANG"	"QIURONG CUI"
[73]	"SEHO PARK"	"SHENG ZHANG"
[75]	"HUIKUN ZHANG"	"ALYSSA DIGILIO"
[77]	"YUQING XU"	"STEPHEN BERG"
[79]	"PEIGEN ZHOU"	"YAN CHEN"
[81]	"QIONG ZHANG"	"YOUNGDEOK HWANG"
[83]	"HAN CHEN"	"NICHOLAS STEPHEN KEULER"
[85]	"ZHENGXIAO WU"	"YUJIN CHUNG"
[87]	"WENWEN ZHANG"	"PERLA REYES"
[89]	"QI TANG"	"CHIA-CHIEH LIN"
[91]	"JIALE XU"	"YANG ZHAO"
[93]	"XINWEI DENG"	"MING XIE"

[95]	"JINGCI MENG"	"ISMOR FISCHER"
[97]	"JIAJIE CHEN"	"ZUOFENG SHANG"
[99]	"HAO ZHENG"	"XU XU"
[101]	"BIN ZHU"	"QUEFENG LI"
[103]	"YU QIU LIU"	"JUN LI"
[105]	"FAN GAO"	"VICTORIA MANSFIELD"
[107]	"ANQI SHI"	"KYLE HEBERT"
[109]	"XIUYU MA"	"CHENLIANG XU"
[111]	"CHANHAN HSU"	"YUANZHI LI"
[113]	"KAM-WAH TSUI"	"XINYU SONG"
[115]	"TIMOTHY IDOWU"	"XIAO NIE"
[117]	"CHEN CHENG"	"ANDREW LESLIE"
[119]	"YAOGUO XIE"	"KEVIN PACKARD"
[121]	"YI LIU"	"WENZHI CAO"
[123]	"JOSEPH DEUTSCH"	"LAN LUO"
[125]	"RYAN ZEA"	"ROBERT WARDROP"
[127]	"SANGBUM CHOI"	"XIN LI"
[129]	"SEUNGBONG HAN"	"ZHIGUO XIAO"
[131]	"RICHARD A. JOHNSON"	"RUI TANG"
[133]	"IAN BRANSTAD RILEY"	"BO HUANG"
[135]	"HEATHER MARIE BRAZEAU"	"XU HE"
[137]	"GARY HOWARD SCHROEDER"	"JUN ZHANG"
[139]	"CHENXI LI"	"SONA Z SWANSON"
[141]	"YUAN JIANG"	"JUNHEE HAN"
[143]	"LILUN DU"	"XIRAN WANG"
[145]	"ZIFENG ZHAO"	"CUIZE HAN"
[147]	"YING ZHANG"	"JOHN DAVIS"
[149]	"CHEN JING"	"JINGLAN LI"
[151]	"YI CHAI"	"THU LE"
[153]	"ALAN HUANG"	"HAOYANG FAN"
[155]	"BOWEN HU"	"TZU HSIANG HUNG"
[157]	"TUN LEE NG"	"ABIGAIL BENZINE"
[159]	"ALEXANDER COVINGTON"	"CHELSEY GREEN"
[161]	"MICHAEL KUTZLER"	"LILI ZHENG"
[163]	"REBECCA POST"	"LU YANG"
[165]	"RUOSI GUO"	"XUN ZHAO"
[167]	"MIN NIU"	"JIAN WU"
[169]	"WEI ZHENG"	"XINJIE HE"
[171]	"JIE WEI"	"CLAUDIA SOLIS LEMUS"
[173]	"XUEYAO CHEN"	"MANJUSHA KANCHARLA"
[175]	"MITCHELL PAUKNER"	"XIAOWU DAI"
[177]	"SHENG WANG"	"CHENGNING ZHANG"
[179]	"GINA OH"	"LUXI CAO"

[181]	"SONG WANG"	"LAM HO"
[183]	"TIEN VO"	"SHIZHEN WANG"
[185]	"SOHEIL SADEGHI"	"BINGYING XIE"
[187]	"GENG LI"	"JENNIFER BIRSTLER"
[189]	"MARIA KAMENETSKY"	"KYLE HERBET"
[191]	"JIANCHANG HU"	"TRAM TA"
[193]	"YUAN WANG"	"KAZUHIKO SHINKI"
[195]	"SIJIAN WANG"	"CHIEN-WEI CHEN"
[197]	"BEHZAD AALIPUR"	"DEREK BEAN"
[199]	"BEN ADAM HAALAND"	"ELOISA D CHAVAS"
[201]	"YALI WANG"	"VARSHA KULKARNI"
[203]	"SCOTT JOSEPH HETZEL"	"MIN JUNG LEE"
[205]	"FANG FANG"	"KRISTEN CYFFKA"
[207]	"TING-LI LIN"	"BRET LARGET"
[209]	"KATHERINE GOODE"	"ERIK NORDHEIM"
[211]	"NING FAN"	"ANRU ZHANG"
[213]	"GARVESH RASKUTTI"	"KRISHNAKUMAR BALASUBRAMANIAN"
[215]	"YAZHEN WANG"	"CHAOYANG YU"
[217]	"YILIN ZHANG"	"ZHENGJUN ZHANG"
[219]	"XIUFENG SHAO"	"YUAN LI"
[221]	"BRET HANLON"	"LILI LAN"
[223]	"CECILE ANE"	"ZHANG CHUNMING"
[225]	"LEI XU"	"MUHONG GAO"
[227]	"AUGUST JENSEN"	"XIAO GUO"
[229]	"THOMAS G. KURTZ"	"NICHOLAS HENDERSON"
[231]	"CHENSHENG KUANG"	"TAERI UHM"
[233]	"NORBERT BINKIEWICZ"	"KARL ROHE"
[235]	"JUNGWON MUN"	"JOSEP GINEBRA"
[237]	"TAO YU"	"XINXIN YU"
[239]	"HAO ZHOU"	"LIE XIONG"
[241]	"QIAN ZHIGUANG"	"SHANG WU"
[243]	"SHAN LU"	"STEPHEN AARON STANHOPE"
[245]	"LANCINE KONATE"	"GUILHERME VIEIRA NUNES LUDWIG"
[247]	"PAUL SAVARIAPPAN"	"CUICUI QI"
[249]	"KJELL DOKSUM"	"DEREK NORTON"
[251]	"KEEGAN KORTHAUER"	"KAILEI CHEN"
[253]	"THEVAASIINEN CHANDERENG"	"JOHN KANE"
[255]	"SOKOL VAKO"	"QING LI"
[257]	"YOUNG LEE"	"HYUNSEUNG KANG"
[259]	"BEHZAD AALIPUR HAFSHEJANI"	"SHULEI WANG"
[261]	"ZHONGJIE YU"	"LIN QI"
[263]	"JIWEI ZHAO"	"SUNDUZ KELES"
[265]	"SHUYUN YE"	"HUI WANG"

[267]	"KEVIN HASEGAWA ENG"	"XIAODAN WEI"
[269]	"MARIA ROJO"	"RUIFENG XU"
[271]	"YONGJOON KIM"	"NELLIE LAUGHLIN"
[273]	"MICHAEL HOGAN"	"TIMOTHY IDOWU"
[275]	"KARL BROMAN"	"REBECCA KOSCIK"
[277]	"RONGJUN ZHU"	"BI CHENG WU"
[279]	"YUNONG LIN"	"ZHIGENG GENG"
[281]	"XIAOPING FENG"	"SHUANG HUANG"
[283]	"GUANNAN SUN"	"JEE YEON KIM"
[285]	"GREGORY SHINAULT"	"TONGHAI YANG"
[287]	"TIMO SEPPALAINEN"	"BENEDEK VALKO"
[289]	"DAVID GRIFFEATH"	"DAVID ANDERSON"
[291]	"PHILIP WOOD"	"FLORIAN BERTRAND"
[293]	"JASON RICHARD SWANSON"	"ALEXANDER KISELEV"
[295]	"SAMUEL STECHMANN"	"WAI TONG FAN"
[297]	"JUN YIN"	"STEFFEN LEMPP"
[299]	"RUIFANG SONG"	"JAMES D KUELBS"
[301]	"ANATOLE BECK"	"DONGHYUN LEE"
[303]	"ANDREJ ZLATOS"	"DANIELE CAPPELLETTI"
[305]	"MIHAELA IFRIM"	"SEBASTIEN ROCH"
[307]	"SCOTT HOTTOVY"	"JONATHON PETERSON"
[309]	"STEPHEN WAINGER"	"ALEXANDER FISH"
[311]	"GREGORIO MORENO-FLORES"	"SUKHENDU MEHROTRA"
[313]	"DIETRICH UHLENBROCK"	"LEV BORISOV"
[315]	"RICHARD A BRUALDI"	"ARNOLD MILLER"
[317]	"MATTHEW BALLARD"	"ALBRECHT KLEMM"
[319]	"PAUL M TERWILLIGER"	"STEVEN SAM"
[321]	"KEN ONO"	"ROBERT HARRON"
[323]	"YANNAN QIU"	"JOHN WILTSHIRE-GORDON"
[325]	"KYUNGMAH KIM"	"CHRISTOPHER WAGNER"
[327]	"PAUL RATHOUZ"	"STEPHEN WRIGHT"
[329]	"BENJAMIN RECHT"	"MICHAEL FERRIS"
[331]	"JESSE THOMAS HOLZER"	"HUILIN HU"
[333]	"ALBERTO DEL PIA"	"ROBERT R MEYER"
[335]	"SHI JIN"	"MARY LINDSTROM"
[337]	"ADIN-CRISTIAN ANDREI"	"MICHAEL LIOU"
[339]	"CHAOQUN MEI"	"CHRISTINA M. KENDZIORSKI"
[341]	"RONALD GANGNON"	"ROBERT WAYNE GREEN"
[343]	"COLE COOK"	"MURRAY CLAYTON"
[345]	"QI JIANG"	"JUN ZHU"
[347]	"ANNE BRUCKNER"	"TING YE"
[349]	"JUN SHAO"	"HUAIBAO FENG"
[351]	"ZIJIAN NI"	"THOMAS COOK"

[353]	"RICHARD J. CHAPPELL"	"MOO K CHUNG"
[355]	"MICHAEL NEWTON"	"BRIAN YANDELL"
[357]	"MICHELLE HARRIS"	"NORMAN DRAPER"
[359]	"YU MENGANG"	"MING YUAN"
[361]	"FANGFANG WANG"	"JUNHO LEE"
[363]	"GRACE WAHBA"	"JASON P FINE"
[365]	"MICHAEL RENE KOSOROK"	"YAJUAN SI"
[367]	"LU MAO"	"DAVID DEMETS"
[369]	"JAMES ANDERSON"	"EDWARD ERKER"
[371]	"SHUAI CHEN"	"MARI PALTA"
[373]	"GUANHUA CHEN"	"SRIKANTHMADHAVAN ARAVAMUTHAN"
[375]	"NATALIA DE LEON GATTI"	"GUILHERME ROSA"
[377]	"YANBING ZHENG"	"CHRISTINE SORKNESS"
[379]	"MARIAN R FISHER"	

```
# Quantos são
nrow(professores_stat)
```

```
[1] 379
```

Exercício 3

```
# Calcula o GPA de cada oferecimento
query <- "
WITH gpa_oferecimento AS (
  SELECT
    course_offering_uuid,
    (
      CAST(a_count AS REAL) * 4.0 +
      CAST(ab_count AS REAL) * 3.5 +
      CAST(b_count AS REAL) * 3.0 +
      CAST(bc_count AS REAL) * 2.5 +
      CAST(c_count AS REAL) * 2.0 +
      CAST(d_count AS REAL) * 1.0 +
      CAST(f_count AS REAL) * 0.0
    )
  /
  (
    CAST(a_count AS REAL) +
    CAST(ab_count AS REAL) +
```

```

        CAST(b_count AS REAL) +
        CAST(bc_count AS REAL) +
        CAST(c_count AS REAL) +
        CAST(d_count AS REAL) +
        CAST(f_count AS REAL)
    ) AS gpa
FROM grade_distributions
WHERE (
    CAST(a_count AS REAL) +
    CAST(ab_count AS REAL) +
    CAST(b_count AS REAL) +
    CAST(bc_count AS REAL) +
    CAST(c_count AS REAL) +
    CAST(d_count AS REAL) +
    CAST(f_count AS REAL)
) > 0
),
oferecimentos_stat AS (
    SELECT DISTINCT
        g.course_offering_uuid,
        c.number AS disciplina,
        i.id AS instructor_id,
        i.name AS professor,
        g.gpa
    FROM gpa_oferecimento g
    JOIN course_offerings co
        ON g.course_offering_uuid = co.uuid
    JOIN courses c
        ON co.course_uuid = c.uuid
    JOIN subject_memberships sm
        ON co.uuid = sm.course_offering_uuid
    JOIN subjects sub
        ON sm.subject_code = sub.code
    JOIN sections sec
        ON co.uuid = sec.course_offering_uuid
    JOIN teachings t
        ON sec.uuid = t.section_uuid
    JOIN instructors i
        ON t.instructor_id = i.id
    WHERE sub.abbreviation = 'STAT'
)
SELECT * FROM oferecimentos_stat;

```

```

"

df <- dbGetQuery(conn, query)

# GPA médio de cada professor
gpa_professores <- aggregate(
  gpa ~ professor,
  data = df,
  FUN = mean
)

# GPA médio de cada disciplina
gpa_disciplinas <- aggregate(
  gpa ~ disciplina,
  data = df,
  FUN = mean
)

# Estatísticas
professor_mais_dificil <- gpa_professores[
  which.min(gpa_professores$gpa),
]

professor_mais_facil <- gpa_professores[
  which.max(gpa_professores$gpa),
]

disciplina_mais_dificil <- gpa_disciplinas[
  which.min(gpa_disciplinas$gpa),
]

disciplina_mais_facil <- gpa_disciplinas[
  which.max(gpa_disciplinas$gpa),
]

# Tabela final com os resultados
resultado <- data.frame(
  Categoria = c(

```

```

        "Professor mais difícil",
        "Professor mais fácil",
        "Disciplina mais difícil",
        "Disciplina mais fácil"
    ),
    Nome = c(
        professor_mais_dificil$professor,
        professor_mais_facil$professor,
        disciplina_mais_dificil$disciplina,
        disciplina_mais_facil$disciplina
    ),
    GPA_Medio = round(
        c(
            professor_mais_dificil$gpa,
            professor_mais_facil$gpa,
            disciplina_mais_dificil$gpa,
            disciplina_mais_facil$gpa
        ),
        2
    )
)

resultado

```

	Categoria	Nome	GPA_Medio
1	Professor mais difícil	JAMES D KUELBS	2.60
2	Professor mais fácil	GUANHUA CHEN	4.00
3	Disciplina mais difícil	431	2.91
4	Disciplina mais fácil	628	4.00

Exercício 4

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-21 21:20:16 -03"
```